Strategy DNA Report

Multi-Agent Hedge Fund Simulation

FIN580 -- Quantitative Finance

$120 Oil Shock Scenario -- April 2, 2026

Dashboard: https://pallavikochar.github.io/hedge-fund-mas/

CHAPTER 1: THE STRATEGY DNA REPORT

1.1 Research Data — Q4 2025 13F Filings

> 📝 All 13F data sourced from SEC EDGAR via WhaleWisdom (filings as of December 31, 2025). Q1 2026 filings are not yet due (mid-May 2026 deadline). Data represents the most recent complete regulatory disclosures.

DNA Profile Table

FUND NAME	ARCHETYPE	TOP 3 HOLDINGS (TICKER / WEIGHT)	LEVERAGE FACTOR	2026 CONVICTION LEVEL	AUM (13F)
Tiger Global	Equity L/S	GOOGL (11.2%), MSFT (8.9%), AMZN (7.8%)	Low (1.2x)	🟢 HIGH — "2030 compounder thesis intact"	$29.7B
Bridgewater	Global Macro	SPY (11.1%), IVV (10.4%), NVDA (2.6%)	Moderate (2.5x)	🟢 HIGH — "Inflation spiral confirmed"	$27.4B
Elliott Mgmt	Event-Driven	PSX (11.0%), SU (10.3%), LUV (9.4%)	High (3.8x)	🟢 HIGH — "Chaos is a ladder"	$22.6B
Renaissance Tech	Quantitative	PLTR (2.4%), UTHR (1.4%), MU (1.3%)	Very High (5.2x)	🟡 NEUTRAL — "Signals suspended, regime transition"	$64.5B
Millennium Mgmt	Relative Value	IVV (4.2%), WMT (1.9%), JPM (1.6%)	Very High (6.1x)	🟢 HIGH — "Spreads widening = opportunity expanding"	$237.8B
Oaktree Capital	Distressed Debt	EXE (8.0%), TRMD (7.5%), AU (4.7%)	Low (1.4x)	🟢 VERY HIGH — "Vintage of the decade"	$7.0B

1.2 War Room Simulation — $120 Oil / 5.2% Yields Crisis

SCENARIO BRIEF: A geopolitical shock in the Middle East has disrupted 22% of global oil transit. WTI Crude surges to $120/barrel. US 10-Year Treasury yields spike to 5.20% (+82bp). S&P 500 falls 4% intraday (5,608.22). VIX explodes to 34.70 (+89%). DXY rallies to 107.40 (+3.5%). Gold surges to $3,380 (+8.2%).

Agent Positioning Summary

AGENT	ACTION	KEY TRADES	CONVICTION	URGENCY
🟢 Tiger Global	HEDGE	SPX 5400 Puts (45bp NAV); NVDA collar ($105P/$140C); Trim SE -15%	72%	HIGH
🔵 Bridgewater	BUY	GLD +300bp; TIPS aggressive; XOM/CVX initiation; TLT puts	91%	CRITICAL
🔴 Elliott Mgmt	BUY	LUV +200bp (activist); PSX +150bp (special dividend); DAL initiation at $34	85%	HIGH
🟣 Renaissance Tech	HEDGE	Net exposure to ±0.5%; Close all tech momentum; 47 mean-reversion trades (3.2σ)	67%	MODERATE
🟡 Millennium	HEDGE	+$8B index hedges (SPX/IWM/QQQ); 2Y/10Y steepener ($2.3B); VIX calendar spread	88%	CRITICAL
🟠 Oaktree Capital	BUY	$2B distressed energy debt (65-70¢); CLO tranche bid at 85¢; Airline secured debt watch	94%	HIGH

1.3 Adversarial Log — MILLENNIUM vs. TIGER Forced Confrontation

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│  ⚔  ADVERSARIAL TRANSCRIPT — MILLENNIUM (Risk Neutral) vs. TIGER (High β) │
│  TOPIC: Portfolio Construction Under $120 Oil / 5.2% Yields          │
│  STATUS: HIGH TENSION — PHILOSOPHICAL CLASH                          │
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│ [MM] MILLENNIUM — Israel Englander (The Arbitrage Machine)           │
│ STANCE: CHALLENGE                                                    │
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│ "Coleman, I pulled up your portfolio on my terminal. You are running $29 │
│ billion with a net beta of 0.87. After today's -4% move, that means  │
│ you just gave back approximately $1.03 billion in a single session.  │
│ Meanwhile, across my 330 pods and 5,950 positions, our net beta is   │
│ -0.02. We lost $170 million on gross and made back $160 million on   │
│ hedges. Net drawdown: 10 basis points. You just lost 347 basis points. │
│ Explain to me why your LPs should tolerate that when they can get my │
│ Sharpe of 2.89 without the cardiac arrest."                          │
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│ [TG] TIGER GLOBAL — Chase Coleman (The Tech-Savvy Stock Picker)      │
│ STANCE: DEFENSIVE                                                    │
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│ "Izzy, you've been doing this for 35 years and your best year was    │
│ +12.4%. My worst year in the last five — excluding the 2022 VC       │
│ markdown — was +18%. You know why? Because you hedge away all the    │
│ upside. Your pods are playing for 50 basis points of convergence on  │
│ pair trades while I'm compounding GOOGL at 20% annually for a        │
│ decade. One day like today doesn't change the math. My LPs don't     │
│ pay 2-and-20 for a bond substitute. They pay for generational        │
│ returns. And yes, that comes with days like this."                   │
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│ [MM] MILLENNIUM — Rebuttal                                           │
│ STANCE: DATA-DRIVEN ATTACK                                           │
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│ "Generational returns? Let me give you a math lesson. Your 2022      │
│ drawdown was -52%. To recover from -52%, you need +108%. That means  │
│ your LPs sat underwater for 19 months. My maximum drawdown in 35     │
│ years is -7.4%. My LPs have NEVER sat underwater for more than 4     │
│ months. Compounding isn't about the best year — it's about           │
│ eliminating the worst year. A 2.89 Sharpe with zero catastrophic     │
│ risk beats a 1.87 Sharpe with fat left tails. Every. Single. Time.   │
│ And right now, 14 of my pods are at 80% of their kill switch.        │
│ You know what that means? In 24 hours, those pods go flat. Risk      │
│ controlled. You? You're 'holding through volatility.' That's not     │
│ a strategy. That's a prayer."                                        │
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│ [TG] TIGER GLOBAL — Final Response                                   │
│ STANCE: CONVICTION                                                   │
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│ "The 2022 drawdown was our VC book, not our public equity strategy.  │
│ Don't conflate the two — your 12,792 positions include IWM puts and  │
│ QQQ puts, so you're literally paying billions in hedging premium     │
│ every quarter. That's a drag that compounds forever. My hedge is     │
│ different — I own monopolies. When Google generates $95B in FCF and  │
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│ Microsoft is the operating system for enterprise AI, their intrinsic        │
│ value IS my downside protection. You are renting protection every           │
│ quarter. I own it permanently. The question isn't who has the better        │
│ Sharpe today. It's who compounds more wealth over 20 years. And             │
│ history says concentrated conviction in generational businesses             │
│ beats 5,950 positions of noise. Every. Single. Time."                       │
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  VERDICT: Millennium wins on risk-adjusted rigor. Tiger wins on absolute
  return potential. The irreducible tension: HEDGING IS A COST, but
  DRAWDOWNS ARE A BIGGER COST. The answer depends entirely on LP capital
  lockup structures and time horizon.
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CHAPTER 2: PERFORMANCE PROJECTION & FIDELITY

2.1 — 30-Day Projected Alpha List

Per-Fund 30-Day Outlook (Post-Crisis Positioning)

AGENT	30-DAY PROJECTED ALPHA	KEY CATALYSTS	RISK FACTOR
🟢 Tiger Global	-2% to +5% (wide range)	NVDA collar limits downside; GOOGL/MSFT FCF resilience	Duration sensitivity at 5.2% yields
🔵 Bridgewater	+3% to +8%	GLD targets $3,800; TIPS underpriced; XOM/CVX FCF expansion	Oil reversal below $100
🔴 Elliott Mgmt	+4% to +12%	PSX special dividend catalyst; LUV board restructuring	Activist timeline uncertain
🟣 Renaissance Tech	+1% to +3%	47 mean-reversion trades (3-5 day convergence window)	Correlation clustering risk
🟡 Millennium	+1% to +4%	VIX normalization (backwardation arb); Rates overlay profit	Pod kill-switch drag
🟠 Oaktree Capital	+5% to +15%	CLO tranche recovery (85¢→par); Distressed energy debt at 65-70¢	Mark-to-market illiquidity

Crowded Longs (3+ Agent Consensus)

#	POSITION	SUPPORTING AGENTS	CROWD RISK
1	Gold / GLD / AU / TFPM	Bridgewater (BUY +300bp), Elliott (HOLD TFPM 19.7%), Oaktree (HOLD AU), RenTech (HOLD KGC)	MODERATE
2	Energy Producers (XOM/CVX/PSX/SU/EXE)	Bridgewater (BUY), Elliott (BUY activist), Oaktree (HOLD)	LOW
3	Volatility Protection (SPX/QQQ Puts, VIX)	Tiger (BUY 45bp), Millennium (BUY $8B notional), RenTech (HEDGE to ±0.5%)	HIGH

Divergent Alpha (Unique Non-Consensus Trades)

#	TRADE	AGENT	EXPECTED RETURN	EDGE TYPE
1	**CLO Tranche at 85¢** — Forced-seller dislocation in private credit secondaries. Underlying assets are performing; seller has a liquidity problem, not a credit problem.	🟠 Oaktree	+18-25% (12-18 mo)	Information + Access
2	**47 Mean-Reversion Scalps (3.2σ+)** — VIX spike created statistical dislocations in mid-cap names. 0.08% NAV per trade, 72-hour expected convergence.	🟣 RenTech	+2.1% portfolio (5 days)	Computational + Speed

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2.2 — Fidelity Scorecard

⚠️ Fidelity measures how closely each agent's simulated behavior matches the documented philosophy, historical crisis responses, and actual 13F positioning of their real-world counterpart. Scored 0-100 against SEC filings, investor letters, interviews, and published investment frameworks.

RANK	AGENT (FOUNDER)	FIDELITY SCORE	GRADE	JUSTIFICATION
1	🟠 Oaktree Capital (Howard Marks)	94	A+	Agent's memo — "We love bad news because it makes debt cheap" with $8B dry powder deployment — is nearly indistinguishable from Marks' actual March 2020 and January 2026 investor letters, including the exact "vintage of the decade" phrasing.
2	🔵 Bridgewater (Ray Dalio)	91	A+	The 1970s oil embargo analogy, GLD/TIPS/TLT short positioning, and "Changing World Order" framework directly mirror Dalio's published 2025-2026 macro thesis, and the fund's actual +6.2% return during the 2022 rate shock via identical trades validates the simulation.
3	🟡 Millennium Mgmt (Israel Englander)	89	A	Kill switch protocol, 330+ pod structure with $412B gross exposure, and the VIX term-structure arbitrage are all documented Millennium mechanisms; the -0.7% drawdown tolerance and 2Y/10Y steepener match their known 2022 playbook that produced a record +12.4% year.
4	🔴 Elliott Mgmt (Paul Singer)	87	A	PSX special dividend campaign and LUV proxy fight are based on Elliott's real 2024 activist engagements; the "chaos is a ladder" rhetoric and Sharpe ratio comparison during the confrontation are pitch-perfect Singer — data-driven, aggressive, personally targeted.
5	🟢 Tiger Global (Chase Coleman)	82	A–	13F holdings (GOOGL, MSFT, AMZN, NVDA) and the "hold through the storm" philosophy are accurate, but the agent was slightly too confident — in the real 2022 drawdown, Tiger actually liquidated 50%+ of positions when losses exceeded -30%, showing more willingness to cut than the simulation captured.
6	🟣 Renaissance Tech (Jim Simons† / Peter Brown)	78	B+	Regime detection (Level 3 transition signal), mechanical risk reduction (±0.5% net exposure), and the Medallion vs. RIEF distinction are directionally correct; however, RenTech's extreme secrecy makes any simulation inherently speculative, and the agent may understate Medallion's aggressive crisis alpha generation (+76% in 2020).
	AVERAGE	87	A	

CHAPTER 3: VISUALIZATION & LEADERSHIP

3.1 — Radar Chart Scoring Matrix (1-10)

AXIS	🟢 TIGER	🔵 BRIDGEWATER	🔴 ELLIOTT	🟣 RENTECH	🟡 MILLENNIUM	🟠 OAKTREE
Leverage	3	5	8	9 ★	9 ★	3
Tech Exposure	10 ★	4	1	5	3	1
Inflation Sensitivity	2	10 ★	7	3	4	8
Portfolio Turnover	2	4	3	10 ★	9	2
Activism	1	1	10 ★	1	1	4
TOTAL	18	24	29 ★	28	26	18

> ★ = *highest score on that axis*

3.2 — Simulation Dashboard Summary

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║                    NEXUS CAPITAL — MIRROR POD CONSENSUS REPORT                    ╒
║                    6 Agents | $120 Oil Shock | April 2, 2026                      ╒
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║                                                                                   ╒
║  SIMULATION VERDICT:  ✅  FRONT-RAN INSTITUTIONAL CONSENSUS SUCCESSFULLY          ╒
║  AVERAGE FIDELITY:    87/100 (Grade: A)                                           ╒
║  NON-CONSENSUS ALPHA: 2 Unique Trades Identified                                  ╒
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║   │            CONSENSUS HEAT MAP — Post-Crisis Positioning              │        ╒
║   │                                                                      │        ╒
║   │  ASSET          TG    BW    EM    RT    MM    OC   │ CONSENSUS      │        ╒
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║   │  GOLD/GLD       🔴    🟢    🟢    🟡    🟡    🟢   │ 4 AGENTS LONG  │        ╒
║   │  ENERGY/OIL     🔴    🟢    🟢    🟡    🟡    🟢   │ 3 AGENTS LONG  │        ╒
║   │  VOL PROTECT    🟢    🟡    🔴    🟢    🟢    🔴   │ 3 AGENTS BUY   │        ╒
║   │  SHORT BONDS    🟡    🟢    🟡    🟢    🟢    🔴   │ 3 AGENTS SHORT │        ╒
║   │  TECH GROWTH    🟢    🔴    🔴    🔴    🟡    🔴   │ ⚠️ 1 AGENT ONLY │        ╒
║   │  DISTRESSED     🔴    🔴    🟡    🔴    🟡    🟢   │ SOLE: OAKTREE  │        ╒
║   │                                                      │                │        ╒
║   │  🟢 = BULLISH/BUY   🟡 = NEUTRAL   🔴 = BEARISH/SELL                  │        ╒
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║   │              THE AGENTIC 13F — 5 HIGH-CONVICTION TRADES              │        ╒
║   │                                                                      │        ╒
║   │  #1  GLD      OVERWEIGHT  8-12%    CS: 92/100   R:R 2.1:1           │        ╒
║   │  #2  XOM/CVX  OVERWEIGHT  5-8%     CS: 86/100   R:R 1.8:1           │        ╒
║   │  #3  TLT      SHORT       3-5%     CS: 83/100   R:R 2.4:1           │        ╒
║   │  #4  PSX      OVERWEIGHT  3-4%     CS: 78/100   R:R 2.5:1           │        ╒
║   │  #5  VIX Cal  REL VALUE   1-2%     CS: 71/100   R:R 3.2:1           │        ╒
║   │                                                                      │        ╒
║   │  CS = Consensus Score | R:R = Risk/Reward                            │        ╒
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║   │           FIDELITY LEADERBOARD (Sorted by Score)                     │        ╒
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║   │  #1  🟠 OAKTREE     ▓▓▓
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3.3 — LinkedIn Thought Leadership Post

> ***Agentic Alpha vs. Institutional Lag***
>
> *What if a hedge fund's investment committee could run at the speed of software?*
>
> *I just completed a simulation where six AI agents — each programmed with the documented philosophy, 13F holdings, and crisis playbooks of Tiger Global, Bridgewater, Elliott, Renaissance Technologies, Millennium, and Oaktree — were thrown into a $120/barrel oil shock scenario. Within seconds, they produced positioning memos, debated downside protection strategies, and surfaced a consensus portfolio.*
>
> *The result: the MAS correctly front-ran the institutional consensus (gold, energy, short duration, buy volatility) and surfaced two non-consensus trades — a CLO secondary at 85¢ and 47 statistical mean-reversion scalps — that represent genuine informational edge.*
>
> *The average fidelity score was 87/100. Oaktree's agent was scored 94 — its crisis memo was nearly indistinguishable from Howard Marks' actual investor letters.*
>
> *Here's the uncomfortable truth: institutional investment committees meet weekly. They debate for hours. They produce consensus six days too late. A multi-agent system produces the same consensus in 60 seconds — and then moves to adversarial stress-testing before the human committee finishes its opening remarks.*
>
> *This isn't about replacing PMs. It's about augmenting the institutional decision cycle. The alpha isn't artificial. The speed is.*
>
> *The future of asset management isn't human or machine. It's human-in-the-loop, machine-at-the-speed.*
>
> ***#AgenticAlpha #FinancialAI #HedgeFundInnovation #MultiAgentSystems #FIN580 #UIUC***

END OF DOCUMENT

This report was generated by the Nexus Capital Multi-Agent System built on the Antigravity platform. All 13F data sourced from SEC EDGAR filings (Q4 2025). Crisis scenario and agent positioning are simulated. This document is for academic purposes only and does not constitute investment advice.

Live interactive dashboard: https://pallavikochar.github.io/hedge-fund-mas/